

Maths-2 Week-1

L1 Vectors

- why vectors?
- we often encounter data in a table.
- Vectors can be thought as a list. It could be either columns or rows.

Eg. $(2, 4, 6, 7, 8) \rightarrow$ a ^{row} vector/matrix

$\begin{bmatrix} a \\ b \\ c \end{bmatrix} \rightarrow$ a column vector/matrix

- vectors can be used to perform arithmetic operations on lists such as addition on columns or rows.

Eg. $\begin{array}{c|c|c|c} & A & B & \\ \hline R & 3 & 5 & 8 \\ D & 2 & 6 & 8 \end{array} \quad \begin{array}{l} (3, 2) \\ + (5, 6) \end{array}$

$(8, 8)$

$$(150, 30, 35) + (-10, -20, -30) = (140, 10, 5)$$

This is how addition of vectors is done. We add the corresponding entries of the vectors.

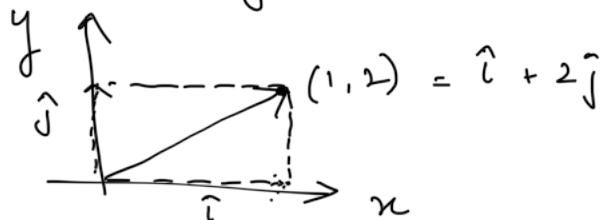
- Scalar multiplication

$$2 \times (2, 2, 4, 6) = (4, 4, 8, 12)$$

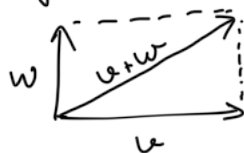
multiplying the entire vector with a constant. We multiply with all the entries of the list.

- Point $(a, b) \equiv$ Vector $(a, b) \equiv a\hat{i} + b\hat{j}$

$\hookrightarrow$ this is how vectors are visualized in $\mathbb{R}^2$.



- by viz method, we can perform vector addition using the parallelogram law.



- Vectors in $\mathbb{R}^n$ are lists (or rows or columns) with n real entries.
 $\hookrightarrow$ or points in $\mathbb{R}^n$.
- Vector has a magnitude & a direction



- exs. of vectors - velocity, force, acceleration.

1.2 Matrices

- A matrix is a rectangular array of nos. arranged in rows & columns

Eg.
$$\begin{bmatrix} 1 & 2 & 3 \\ 3 & 2 & 1 \end{bmatrix}_{2 \times 3}$$

$\downarrow$ row $\downarrow$ column

(i, j) -th entry of a matrix occurring in the i -th row & j -th column.

$(1, 2)$ -th entry $\rightarrow 2$

- A sq. matrix has equal no. of rows & columns.

$$\begin{bmatrix} - & - \\ - & - \end{bmatrix}_{2 \times 2}$$

i -th diagonal entry of a sq. matrix is (i, i) -th entry.

The diagonal of a sq. matrix is the set of diagonal entries.

- The sq. matrix with all entries except for diagonal are 0 is c/d a diagonal matrix.

$$\begin{bmatrix} - & 0 \\ 0 & - \end{bmatrix}$$

- Scalar matrix - diagonal matrix + all diagonal entries are equal.

$$\begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} = 2$$

- Identity matrix - scalar matrix + all diagonal entries is 1.

$$\underline{I} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = 1$$

- Set of linear eq^{ns} can be represent as matrices.

$$\begin{array}{l} 3x + 4y = 5 \\ 4x + 6y = 10 \end{array} \Rightarrow \left[\begin{array}{cc|c} 3 & 4 & 5 \\ 4 & 6 & 10 \end{array} \right]$$

- Addⁿ of matrix $\Rightarrow$ same no. of rows & column. Add of corresponding entries.

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} + \begin{bmatrix} 2 & 1 \\ 1 & 3 \end{bmatrix} = \begin{bmatrix} 3 & 1 \\ 1 & 4 \end{bmatrix}$$

$$\boxed{(A+B)_{ij} = A_{ij} + B_{ij}}$$

- Scalar multiplicatⁿ - multiply the matrix with a constant

$$2 \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} = \begin{bmatrix} 2 & 4 \\ 6 & 8 \end{bmatrix}$$

$$\boxed{(cA)_{ij} = c(A_{ij})}$$

- Matrix multiplicatⁿ

$$\text{eg. } \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}_{2 \times 2} \times \begin{bmatrix} 1 & 2 & 3 \\ 3 & 4 & 5 \end{bmatrix}_{2 \times 3} = \begin{bmatrix} 7 & 10 & 13 \\ 15 & 22 & 29 \end{bmatrix}_{2 \times 3}$$

Eg. $\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}_{2 \times 2} \times \begin{bmatrix} 5 \\ 6 \end{bmatrix}_{2 \times 1} = \begin{bmatrix} 17 \\ 39 \end{bmatrix}_{2 \times 1}$

$$A_{m \times n} \times B_{n \times p} = C_{m \times p}$$

no. of columns in the first matrix is equal to the no. of rows in second matrix.

$$(AB)_{ij} = \sum_{k=1}^n A_{ik} B_{kj}$$

Eg. $\begin{bmatrix} c & 0 & 0 \\ 0 & c & 0 \\ 0 & 0 & c \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{bmatrix} = c \begin{bmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{bmatrix}$

- Properties

1. Associative - $(A+B) + C = A + (B+C)$

2. " - $(AB)C = A(BC)$

3. $A+B = B+A$ (commutative)

4. $AB \neq BA$ (not necessarily)

5. $\lambda(A+B) = \lambda A + \lambda B$

6. $\lambda(AB) = (\lambda A)B = A(\lambda B)$

7. $A(B+C) = AB + AC$

8. $(A+B)C = AC + BC$

L3 System of Linear Equations

	A	B	C
R	8	12	3
D	8	5	2
U	4	7	5

1960 2215 1135

$R \rightarrow x$

$D \rightarrow y$

$U \rightarrow z$

$$8x + 12y + 3z = 1960$$

$$8x + 5y + 2z = 2215$$

$$3x + 7y + 5z = 1135$$

We can obviously solve these eqns to solve for x, y, z and calculate cost of R, D & O.

- A linear eqn is an eqn of form -

$$a_1x_1 + a_2x_2 + \dots + a_nx_n = b$$

$a_s \rightarrow$ coefficients

$x_s \rightarrow$ variables

- A sys. of linear eqn is collecⁿ of several linear eqn. that involves same set of variables.

- A solⁿ of such sys. such that all the eqns are simultaneously satisfied.

- The general sys. has m eqns and n unknown variables

- how eqs to matrix $\Rightarrow \boxed{Ax = b}$

$A_{m \times n}$
 $\Downarrow$
 coeff.
~~matrix~~
 matrix

$x_{n \times 1}$
 $\Downarrow$
 column
 matrix
 with
 unknown
 vars.

$= b_{m \times 1}$
 $\Downarrow$
 constant
 matrix
 with values
 of R.H.S.

$\nearrow$ this is
 x matrix

Eg. $1x + 2y + 3z = 4$
 $3x + 4y + 5z = 6$

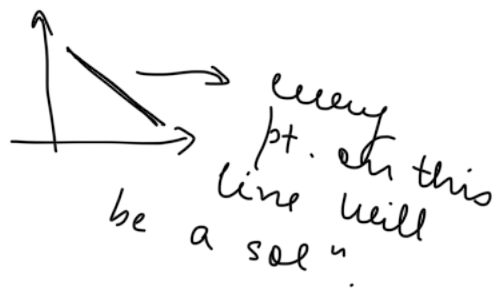
$$\begin{bmatrix} 1 & 2 & 3 \\ 3 & 4 & 5 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 4 \\ 6 \end{bmatrix}$$

- There are 3 possibilities in terms of solⁿ for sys. of linear eqⁿ

1. ∞ solⁿ
2. single solⁿ
3. no solⁿ.

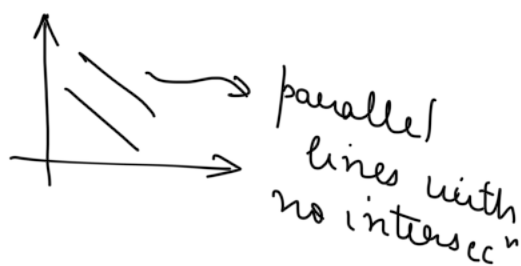
Eq. ∞ solⁿ

$$\begin{aligned} 2x + y &= 215 \\ 4x + 2y &= 430 \end{aligned}$$



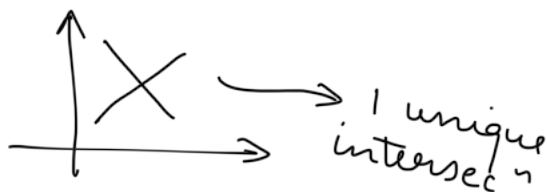
Eq. no solⁿ

$$\begin{aligned} 2x + y &= 215 \\ 4x + 2y &= 400 \end{aligned}$$



Eq. 1 solⁿ

$$\begin{aligned} 2x + y &= 215 \\ 3x + y &= 460 \end{aligned}$$



LY Determinants (Part 1)

- every sq. matrix A has an ass. no. c/d determinant and given by $\det(A)$ and /or $|A|$.

- $A = [a]$

$$\det(A) = a$$

- $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$

$$\boxed{\det(A) = ad - bc}$$

- Eq. $A = \begin{bmatrix} 2 & 3 \\ 6 & 10 \end{bmatrix} = 20 - 18 = 2$

$$A = \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix}$$

$$\begin{aligned} \det(A) &= a \begin{bmatrix} e & f \\ h & i \end{bmatrix} - b \begin{bmatrix} d & f \\ g & i \end{bmatrix} + c \begin{bmatrix} d & e \\ g & h \end{bmatrix} \\ &= a(ei - hf) - b(di - gf) \\ &\quad + c(dh - ge) \end{aligned}$$

Eg. $A = \begin{bmatrix} 2 & 4 & 1 \\ 3 & 8 & 7 \\ 5 & 6 & 9 \end{bmatrix}$

$$\begin{aligned} \det(A) &= 2(72 - 42) - 4(27 - 35) + \\ &\quad 1(18 - 40) \\ &= 2(30) - 4(-8) + 1(-22) \\ &= 60 + 32 - 22 = 70 \end{aligned}$$

- det of identity matrix is always 1.

- det of product of matrices

$$\boxed{\det(AB) = \det(A) \det(B)}$$

- $AA^{-1} = I = A^{-1}A$

$$\det(A) \det(A^{-1}) = 1$$

$$\det(A^{-1}) = \frac{1}{\det(A)}$$

$$\boxed{\det(A^{-1}) = \det(A)^{-1}}$$

- if we switch 2 rows / 2 columns of a matrix -

$$\boxed{\det(A') = -\det(A)}$$

- if we add multiple of a row to

another row / we do similar opⁿ to a column.

$$\boxed{\det(A') = \det(A)}$$

- if we perform scalar multiplication of a row / column by a constant t :

$$\boxed{\det(A') = t \det(A)}$$

LS Determinants Part - 2

- we have studied how to calculate the determinant till 3×3 matrix.

$$A = \begin{bmatrix} 2 & 4 & 3 \\ 0 & 8 & 7 \\ 0 & 0 & 9 \end{bmatrix}$$

$$\det[A] = 2 \times 8 \times 9$$

$$= 16 \times 9$$

$$= 144$$

↑
upper triangular matrix

↓
product

$(A_{ij} = 0, \text{ if } i > j)$ $\leftrightarrow$ of diagonal entries
lower Δ matrix

$$(A_{ij} = 0, \text{ if } i < j)$$

- The Transpose of $A_{m \times n}$ is the $n \times m$ matrix with (i, j) entry A_{ji} .

$$\boxed{(A^T)_{ij} = A_{ji}}$$

$$\boxed{\det(A^T) = \det(A)}$$

- A is $n \times n$ sq., $n \leq 4$. Minor of the entry in the i^{th} row & j^{th} column is the determinant of sub-matrix formed by deleting i^{th} row & j^{th} column.

$$A = \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix}$$

$$M_{11} = \det \begin{bmatrix} e & f \\ h & i \end{bmatrix}$$

$(i, j)^{\text{th}}$ cofactor

$$C_{ij} = (-1)^{i+j} M_{ij}$$

$$- \det(A) = a_{11} \times M_{11} - a_{12} \times M_{12} + a_{13} \times M_{13}$$

$$= a_{11} \times C_{11} + a_{12} \times C_{12} + a_{13} \times C_{13}$$

- we can generalize our results beyond $n=3$. Use the inductive effect.

$$\boxed{\det(A) = \sum_{j=1}^n a_{1j} C_{1j}}$$

$$\text{where } C_{ij} = (-1)^{i+j} M_{ij}.$$